TUTORIAL-9

training

(20) datapts → 5 fold K-fold CV $\frac{20}{5} = 4$

Training (16) | validation (4)

KNN (K=3)

Fold-1 (16, 4)

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(12) datapoints → 4 fold CV $\frac{12}{4} = 3$

9 | validation (3)

Fold-1:
KNN (K=3).

(9, 3) | (25, 50, 8)

$d = \sqrt{(16)^2 + (47)^2 + 8^2}$

$= \sqrt{256 + 2209 + 64}$

$= \sqrt{2529} = 50.2$

d =

validation

10 → (46, 84, 3) ⇒ N as majority of nearest is N

(25, 50, 8)

9

$d = \sqrt{(21)^2 + (34)^2 + (-5)^2}$

$= \sqrt{441 + 1156 + 25}$

$= \sqrt{1622} = 40.24$

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ii) (45, 80, 3) & (46, 84, 3)

$d = \sqrt{(1)^2 + (4)^2 + 0^2} = \sqrt{1+16} = \sqrt{17} = 4.123$ - N

iii) (28, 55, 9) & (46, 84, 3)

$18^2 + 29^2 + 6^2 =$

$324 + 841 + 36$

$\sqrt{1201} = 34.6$

iv (50, 85, 2) & (46, 84, 3)

$4^2 + 1^2 + (-1)^2 = \sqrt{16+1+1} = 4.24$ - N

v) (30, 60, 10) & (46, 84, 3)

$16^2 + 24^2 + 7^2 = 256 + 576 + 49$

$= \sqrt{881} = 29.6$ - Y.

vi) (48, 82, 3) & (46, 84, 3)

$= 2^2 + 2^2 + 0^2 = \sqrt{4+4+0} = 2.8$ - N

vii) (26, 52, 8) & (46, 84, 3)

$= (20)^2 + (32)^2 + 5^2 = 400 + 1024 + 25$

$= \sqrt{1449} = 38.06$

viii) (47, 88, 2) & (46, 84, 3)

$= 1^2 + 4^2 + 1^2 = \sqrt{1+16+1} = 4.24$ - N

ix) (29, 58, 9) & (46, 84, 3)

$= 17^2 + 26^2 + 6^2 = 289 + 676 + 36$

$= \sqrt{1001} = \cancel{31.6}$ 31.6

~~4.123 → N~~

~~4.24~~ N

Next validation set (27, 54, 8) – Y

(27, 54, 8) & (46, 84, 3)

$19^2 + 30^2 + 5^2 = 361 + 900 + 25 = \sqrt{1286}$

$= 35.86$

(27, 54, 8) & (25, 50, 8)

$= 2^2 + 4^2 + 0^2 = \sqrt{20} = 4.49$ – Y

(27, 54, 8) & (45, 80, 3)

$= 32.01$

(27, 54, 8) & (28, 55, 9)

$=$ ~~7.141~~ 1.713 – Y

(27, 54, 8) & (50, 85, 2)

$= 39.16$

(27, 54, 8) & (30, 60, 10)

$= 7$ – Y

(27, 54, 8) & (48, 82, 3)

$= 36.35$

(27, 54, 8) & (26, 52, 8) – 2.23 – Y

(47, 88, 2) – 39.89

(29, 58, 9) – 4.58 – Y

$2^2 + 4^2 + 1^2$

Non-validati (49, 86, 2) - N

(49, 86, 2) (25, 50, 8) → 43.68

(45, 80, 3) → 7.28 - N

(28, 55, 9) → 38.09

(50, 85, 2) → 1.41 - N

(30, 60, 10) → 32.18 - Y

(48, 82, 3) → 4.24 - N

(26, 52, 8) → 41.48

(47, 88, 2) → 2.82 - N

(29, 58, 9) → 35.11

Accuracy → 100%